

Preetom Kumar Biswas

Ph.D Student,
Computer Science and Engineering
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Experience

- a. **Graduate Student Research Assistant (UMich)** September 2026 - Present
Supervisor: Prof. Alexander Rodriguez,
Department of Computer Science & Engineering, University of Michigan, Ann Arbor, MI, USA
Research Interests: Timeseries Analysis, Causal Inference, Scenario generation, Intelligent Multi-Agent Systems, Diffusion Models & LLMs, AI for Science.
- b. **Graduate Research Assistant (ASU)** December 2025 - May 2026
Supervisor: Prof. Giulia Pedrielli, School of Computing & Augmented Intelligence, Arizona State University, USA
Topic: Denoising Diffusion Probabilistic Models (DDPM), Scenario Generation, LLMs, Causal Inference, Multi-variate Time Series
Thesis Overview:
 - Exploring falsified and ethically complex scenarios expressed as **Signal Temporal Logic (STL)** sequences generation using generative models such as **Denoising Diffusion Probabilistic Models (DDPM)** for autonomous driving systems.
 - Developing **LLM-driven rule-based causal inference framework** to extract explainable and reproducible logical relations for multi-variate time-series signals.
- c. **Graduate Teaching Assistant (ASU)** August 2025 - December 2025
Courses: FSE100 (Introduction to Engineering), Prof. Steven Osburn
Responsibilities: Collaborating with Professor to design diverse learning material for 3 sections with 100+ students, holding classes and teaching various technologies such as MATLAB, Javascript, Signal Processing, and Engineering principles.
- d. **Undergraduate Research Assistant (ASU)** August 2023 - August 2025
Supervisor: Prof. Giulia Pedrielli, School of Computing & Augmented Intelligence, Arizona State University, USA
Topic: Extended SEIR model simulation, Non-linear Optimization, Agent-Based Modeling, Evolutionary Game Theory, Causal Inference
Contributions:
 - Modeled SEIR based Py-SIRTEM model to predict and simulate temporal evolution of COVID-19 transmission that provides data on 76 different medical states.
 - Applied **Sequential Monte Carlo Methods (ParticleFilters)** and **Bayesian Optimization** to learn and validate intrinsic model hyperparameters such as transmission and infection rates
 - Devised an **Agent-Based Adaptive model** to simulate individual behavior and optimize government and economic policies based on **evolutionary game theory** principles.
- e. **Undergraduate Research Intern (Remote)** August 2020 - August 2024
Supervisors: Prof. Preetam Ghosh, Virginia Commonwealth University, USA and Prof. Satyaki Roy, University of Alabama in Huntsville, USA
Topic: SEIR Modeling, constraint optimization, gene regulatory networks, influence diffusion
Contributions:
 - Implemented machine learning and optimization algorithms using Python libraries (scipy, scikit-learn) to model spatiotemporal contagion evolution and optimal intervention strategies during pandemic.
 - applied the **network inference algorithm, GENIE3**, to infer the influence diffusion and spatiotemporal evolution of COVID-19 virus - the results of which were published in *Scientific Reports* in 2021
 - Employed multithreading and GPU computing to improve program and simulation efficiency.

- f. **Undergraduate Teaching Assistant (ASU)** August 2022 - May 2025
Courses: CSE110 (Principles of Programming) & CSE205 (OOP & Data Structures) & CSE310 (Data Structures & Algorithms) & CSE360 (Intro to Software Engineering) & CSE471 (Intro to AI)
Responsibilities: Collaborating with Professor to design diverse learning material, holding regular office hours & pre-exam review sessions
- g. **Summer Research Intern 2024** May 2024 - July 2024
Supervisors: Prof. YooJung Choi, School of Computing & Augmented Intelligence, Arizona State University, USA
Topic: Probabilistic Circuits, Bayesian inference, expectation maximization, sampling techniques
Contributions:
- Learned and trained intrinsic Hidden Chow-Liu Tree structure of **Probabilistic Circuits (PCs)** using latent feature vectors of processed images from a pre-trained **Vector Quantized Generative Adversarial Network (VQGAN) encoder-decoder model** using joint log likelihood maximization
 - Programmed conditional query operations such as (MAP, Sampling, Sampling with MAP at leaves) on PCs using Julia to infer and generate latent feature vectors of images.
 - Calculated accuracy and FID (Fréchet Inception Distance) score to gauge quality of the generated images and to optimize the circuit parameters.
- h. **Summer Research Intern 2023** May 2023 - July 2023
Supervisors: Prof. Giulia Pedrielli, School of Computing & Augmented Intelligence, Arizona State University, USA
Topic: Extended SEIR model simulation, Non-linear Optimization, Code refactorization
Contributions:
- Refactored existing MATLAB code to improve performance and usability of single-city model by 50% by switching to Python, used ddeint library to solve delay differential equations, designed autoregressive algorithms to compute infection rates based on user specifications for the model
 - Constructed dashboard app for Virus Simulation Model (Py-SIRTEM) using Python dash library to facilitate wider usability and distribution.
- i. **Senior Fullstack Developer** April 2023 - May 2024
Attendance Management System project, sponsored by Arizona State University
Contributions:
- Designed the frontend for the fully automated and auto graded attendance management system at ASU, used typescript and react.js, managed the frontend team, delegated weekly deliverables
 - Incorporated API requests to display frontend using processed data from card scanners and auto update grades on Canvas LMS.
- j. **Technology Officer** April 2023 - May 2024
Combat Ready Robotics Club, Arizona State University
Contributions:
- Held workshops to teach students about design and electronics relevant to building battlebots
 - Developed Tournament Management website, BotBracket, for club's annual battlebots event using MERN stack, automated scoring and match history, reduced operation time by 75
 - Reorganized club website to display latest event info, managed meeting schedule and notifications through discord bot

Publications: Peer-reviewed Journals [\[Google Scholar\]](#)

- Tanmay Khandait, **Preetom Biswas**, Hideki Okamoto, Bardh Hoxha, Georgios Fainekos, and Giulia Pedrielli, Diffusion-Guided Search via Exponential Tilting (DiffTilt): An Application to Falsification of Safety-Critical Systems, *International Conference on Embedded Software (EMSOFT)*, 2026 (Accepted)
- Yahia Alqurnawi, **Preetom Biswas**, Anmol Rao, Tejas Anvekar, Chitta Baral, Vivek Gupta, ViTaB-A: Evaluating Multimodal Large Language Models on Visual Table Attribution, *Efficient Spatial Reasoning Workshop @ ICLR*, 2026 (Accepted)

- **Preetom Biswas**, Giulia Pedrielli, and K. Selçuk Candan, Causality by Abstraction: Symbolic Rule Learning in Multivariate Timeseries with Large Language Models, *3rd International Conference on Neuro-Symbolic Systems (NeuS)*, 2026 (Accepted)
- **Preetom Biswas**, Giulia Pedrielli, and K. Selçuk Candan, PySIRTEM: An Efficient Modular Simulation Platform for the Analysis of Pandemic Scenarios, In *Proceedings of the 2025 Winter Simulation Conference* (Published)
- Satyaki Roy, **Preetom Biswas**, and Preetam Ghosh, Advancing Infection Profiling under Data Uncertainty through Contagion Potential, *PLOS ONE*, 2025 (Published)
- Satyaki Roy, **Preetom Biswas**, and Preetam Ghosh, Determining the rate of infectious disease testing through contagion potential, *PLOS Global Public Health*, 2023 (Published)
- Satyaki Roy, **Preetom Biswas**, and Preetam Ghosh, Spatiotemporal Tracing of Pandemic Spread from Infection Data, *Scientific Reports*, 2021 (Published)
- Satyaki Roy, **Preetom Biswas**, and Preetam Ghosh, Quantifying Mobility and Mixing Propensity in the Spatiotemporal Context of a Pandemic Spread, *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2021 (Published)
- Satyaki Roy, **Preetom Biswas**, and Preetam Ghosh, Effectiveness of Network Interdiction Strategies to Limit Contagion During a Pandemic, *IEEE Access*, 2021 (Published)

Projects

- **SCAI GPT - Instructional Assistance System:** An exclusive ASU conversational GPT finetuned on specific canvas course content and regularly updated through discussion board posts to provide guided help to students with class materials without giving away explicit answer. Frameworks: Flask backend, vite react frontend, PostgreSQL and Qdrant vector database.
 - Utilized Canvas LMS API to extract course discussions, pdfs and documents from courses to create RAG pipeline for accurate course-specific queries
 - Employed LangChain and semantic chunking to process and embed the materials before storing vector representations in a QDrant database to enable efficient semantic search
 - Embedded the system into Canvas using Google OAuth and LTI (Learning Tools Interoperability) to improve security and user convenience.
 - Applied DeepEval to benchmark performance finding retrieval accuracy of 90% and BLEU score of 0.87
- **BotBracket:** A tournament management system web application using MERN (MongoDB, Express.js, React.js, Node.js) stack to automate BattleBot match setup, score update and tournament data storage. Other functionalities include user authentication and live updates. [\[Link\]](#)
- **Attendance Management System AMS:** ASU sponsored Learning Management application to streamline attendance in classes using card scanners and Canvas LTI integration, with Java springboot backend, NextJS frontend, and PostgreSQL database server.
- **Project Spyn:** An automated robot using MATLAB and Lego Mindstorms EV3 kit that can navigate mazes and safely transports passengers (FSE100 final project).

Educational Qualifications

- Computer Science & Engineering (Ph.D),** Sept 2026 - Present
University of Michigan
 Ann Arbor, MI, USA
 [GSRA, Fall 2026 and MICDE Fellow 2026-2027]
- Computer Science (MS),** Ira A. Fulton Schools of Engineering, Aug 2025 - May 2026
Arizona State University (ASU)
 Tempe, AZ, USA [Graduate TA, Fall 2025 and Graduate RA, Spring 2026]
 Current GPA: **3.92**
- Computer Science (BS),** Ira A. Fulton Schools of Engineering, Aug 2021 - May 2025
Arizona State University (ASU)
 Tempe, AZ, USA [Recipient of New American University Award]
 Current GPA: **4.00**

- d. **Bachelor of Science (B.Sc.)**, Computer Science and Engineering Feb 2020 - Feb 2021
Bangladesh University of Engineering and Technology (BUET)
Dhaka, Bangladesh
- e. **Higher Secondary School Certificate (H.S.C)**, Science Division July 2019
Govt. M. M. College, Jessore, Bangladesh
Ranked 1st [out of ~ 130,000 students] in Jessore Education Board (One of 9 boards in Bangladesh)
Cum. GPA: 5.0 out of 5.0
Marks: 1232 out of 1300
- f. **Secondary School Certificate (S.S.C)**, Science Division May 2017
Police Line Secondary School, Jessore, Bangladesh
Ranked 23rd [out of ~ 200,000 students] in Jessore Education Board
Cum. GPA: 5.0 out of 5.0
Marks: 1211 out of 1300
- g. **Junior School Certificate (J.S.C)** December 2014
Police Line Secondary School, Jessore, Bangladesh
Ranked 2nd [out of ~ 225,000 students] in Jessore Education Board
Cum. GPA: 5.0 out of 5.0

Coursework

• **CSE576**: Natural Language Processing • **CSE598**: Bio-driven AI and optimization • **CSE310**: Data Structures and Algorithms • **CSE471**: Introduction to Artificial Intelligence • **CSE472**: Social Media Mining • **CSE475**: Introduction to Machine Learning • **CSE568**: Biocomputing • **MAT343**: Applied Linear Algebra • **PHY121**, **PHY131**: Mechanics, Electromagnetism

Scholarships and Achievements

- Received **Outstanding UGTA & Legacy UGTA Awards** for 3 years of service as Undergraduate Teaching Assistant, **Fulton School of Engineering, 2025**, Arizona State University, Tempe, AZ, USA
- Received **Best Use of Git Award, SunHacks, 2023**, Arizona State University, Tempe, AZ, USA
- Ranked 4th [against BS, MS, & PhD competitors], **SoDA Code Challenge IX, 2023**, Arizona State University, Tempe, AZ, USA
- **Finalist & Most Destructive Bot** at **SunDevil Smackdown, 2023**, Combat Ready Robotics Club, Arizona State University, Tempe, AZ, USA
- Ranked 3rd [against BS, MS, & PhD competitors], **WiCS 17th Annual Programming Competition, 2022**, Arizona State University, Tempe, AZ, USA
- Received **New American University Award** (4 years) [Top Scholarship for International undergraduates], Arizona State University, Tempe, AZ, USA
- Ranked 17th [out of ~ 1 million students], Bangladesh University of Engineering and Technology (BUET) entrance exam 2019
- Ranked 5th [out of ~ 1 million students], Dhaka University entrance exam 2019
- Ranked 21st [out of ~ 1 million students], Khulna University of Engineering and Technology (KUET) entrance exam 2019
- Ranked 1st [out of ~ 130,000 students], Higher Secondary School Certificate (H.S.C), Jessore Board 2019
- Ranked 23rd [out of ~ 200,000 students], Secondary School Certificate (S.S.C), Jessore Board 2017
- Ranked 2nd [out of ~ 225,000 students], Junior School Certificate (J.S.C), Jessore Board 2014
- Champion, Regional Math Olympiad 2018
- Qualified for the National Spelling Bee (English) 2014
- Ranked 3rd, Divisional 'English in School' Contest 2014

Skills

- **Programming Languages** : Java, Python, C/C++, Lua, MATLAB, Scheme, Prolog, Swift, Julia

- **Front-end & Back-end** : HTML, CSS, Sass, JavaScript, Typescript, React.js, Node.js, Express.js, MongoDB, SQL
- **Tools** : LaTeX, SolidWorks, Git
- **Languages** : Bangla, English, Hindi

Test Scores

- **GRE**: 329/340
 - **Verbal**: 161/170
 - **Quantitative**: 168/170
 - **Analytical Writing**: 4/6
- **SAT**: 1500/1600
 - **Math**: 780/800
 - **English**: 720/800
- **SAT Subject Tests**: 2400/2400
 - **Physics**: 800/800
 - **Chemistry**: 800/800
 - **Math 2**: 800/800
- **IELTS**: 8.0
 - **Listening**: 8.5
 - **Reading**: 8.0
 - **Speaking**: 8.0
 - **Writing**: 7.5

Extra-curricular Activity

- Regular Participant in SunHacks, SoDA and WiCS Programming competitions at ASU
- Participant in Battlebot style robotics tournaments by Combat Ready Robotics at ASU
- Participant in Regional and National Math Olympiad
- Participant in Regional, Divisional and National Spelling Bee (English)
- Participant in Regional Physics Olympiad
- Participant in Regional and Divisional 'English in School' contests
- Participant in Regional and Divisional 'Srijonshil Medha Anweshan' (Creative Talent Hunt)
- Occasional writer in high school magazine 'Barshiki'

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